

Calls and songs

Nature's airwaves are crowded with animal sounds, and so are the oceans and seas. Small animals can be surprisingly loud, and whale songs can reach more than 180 decibels—even louder than a passenger jet. Many animals make sounds by breathing out so that the air vibrates, but some insects rub body parts together or click membranes that work like tiny high-pitched drums. Sounds help animals to attract a mate and let them keep in touch, sometimes over astounding distances. Sounds can also keep danger at bay. In emergencies, a warning hiss can stop a predator in its tracks.



One among many

Northern gannets nest in densely packed colonies containing up to 50,000 breeding pairs and their young. Amazingly, they use sound to recognize one another, despite the clamor of all the other birds and the background noise of the wind and the waves. Adults honk as they come in to land, and their partners need less than a second to identify them. They call back, guiding their partner to touch down in the right place.

Practice makes perfect

A sedge wren weighs less than ½ oz (10 g), but its chattering song carries a long way in still air. Like all songbirds, it has a well-developed voicebox, or syrinx, deep inside its chest. The syrinx contains a complex set of muscles and membranes, which tighten and relax to produce a fast-changing stream of notes. Male wrens are born with the instinct to sing, but they learn their song by listening to and copying the older males.

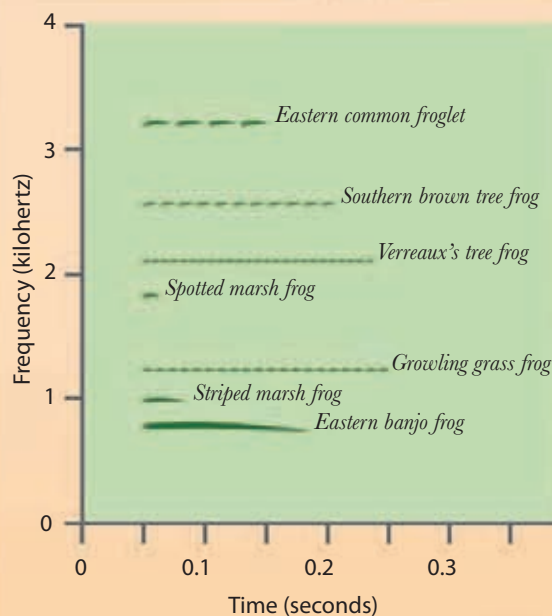


Outside broadcast

In spring and summer, male green tree frogs croak to attract a mate. They have vocal cords, like we do, but they also have a vocal sac, which works like an inflatable amplifier. To start calling, a frog breathes in and then closes its nostrils. It forces the air backward and forward between its lungs and vocal sac, so that its vocal cords can make the air vibrate. Green tree frogs have a single sac, but many other frogs have one on each side of their heads.



Air resonates inside vocal sac, amplifying frog's call



Private audience

The males of each species of frog have their own distinctive call, making sure that they attract females of the right species. This is particularly important when several species breed in the same place. The calls of seven Australian frogs are shown on this graph. They often use the same pools to breed in and call at the same time of day, but each call has its own frequency, or pitch, and its own rhythm and timing.